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# Stuffed toys as a source of oral PFAS exposure: Migration into artificial saliva and exposure assessment

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**BACKGROUND:** Per- and polyfluoroalkyl substances (PFAS) are widely used in textiles and consumer products, resulting in ubiquitous human exposure. Infants and young children may experience distinct exposure patterns due to mouthing behavior, yet oral exposure from textile-based products such as stuffed toys remains poorly characterized.
**OBJECTIVE:** This study aimed to quantify saliva-mediated PFAS migration from commercially available stuffed toys under mouthing-relevant conditions, assess the potential contribution to early-life exposure, and evaluate the effectiveness of washing as a mitigation measure.
**METHODS:** Eighteen stuffed toys intended for infants and toddlers were analysed. PFAS migration was assessed using artificial saliva under standardized conditions and quantified by Liquid Chromatography-Tandem Mass Spectrometry (LC–MS/MS). Deterministic exposure assessment was performed to estimate daily and weekly intake, complemented by margin of exposure (MOE) calculations and reverse dosimetry based on biomonitoring data. The effect of a standardized water-rinsing step on PFAS migration was also investigated.
**RESULTS:** PFAS were detected in all toys, with migration profiles dominated by short-chain perfluoroalkyl carboxylic acids, particularly perfluorohexanoic acid (PFHxA). A limited number of toys accounted for most of the PFAS released. Estimated intakes for EFSA-regulated PFAS (perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorohexanesulfonic acid (PFHxS), perfluorononanoic acid (PFNA)) remained below the tolerable weekly intake (TWI), whereas MOE and reverse-dosimetry analyses highlighted compound- and product-specific exposure relevance for short-chain and emerging PFAS. Washing reduced PFAS migration in most cases but increased migration for some PFAS and products, indicating heterogeneous and non-systematic effects.
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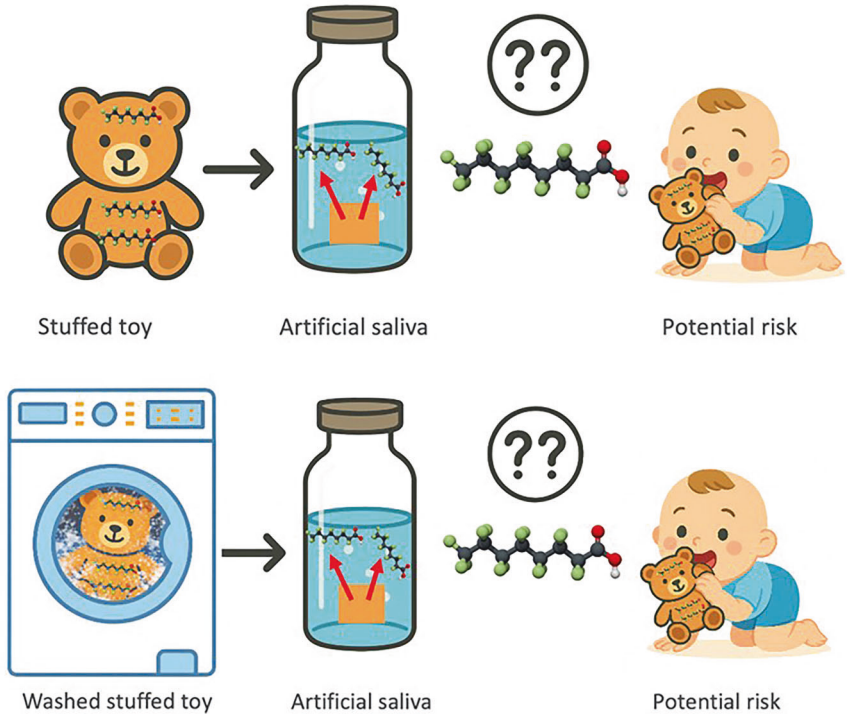
- This study demonstrates that stuffed toys represent a relevant and often underappreciated source of early-life PFAS exposure. PFAS migration was observed across all products and was dominated by short-chain compounds, with perfluorohexanoic acid (PFHxA) emerging as the main driver of exposure relevance. While intake of EFSA-regulated PFAS remained below guidance values, PFHxA showed a large influence on exposure metrics, including reduced MOE and high relative contributions in reverse-dosimetry analyses. These findings reveal an important regulatory and knowledge gap for emerging PFAS and highlight the need to integrate mouthing-relevant exposure pathways into risk assessment of products intended for young children.

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**Graphical Abstract**




## INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) constitute a large and heterogeneous class of synthetic fluorinated chemicals that have been widely used since the 1950s due to their water-, oil- and stain-repellent properties [1]. Their remarkable resistance to chemical, thermal and biological degradation stems from the strength of the carbon–fluorine bond, earning PFAS the designation of "forever chemicals" [2]. More than 4'700 PFAS are currently listed in the inventory of the Organisation for Economic Co-operation and Development / United Nations Environment Programme (OECD/UNEP), while broader structural definitions encompass tens of thousands of fluorinated substances, underscoring the complexity of their environmental behavior and toxicological assessment [1, 2]. Decades of industrial production and widespread consumer use have resulted in global PFAS contamination across all major environmental compartments, including air, soils, sediments, surface and groundwater, as well as precipitation and remote regions reached through long-range atmospheric and oceanic transport [3, 4]. Consequently, human exposure is virtually unavoidable and arises from multiple, often additive, pathways, including contaminated drinking water, dietary intake, indoor air and dust, and direct contact with PFAS-treated consumer products [5, 6].

Once absorbed, many PFAS bind strongly to serum proteins and undergo active renal reabsorption, resulting in biological half-lives of several years for legacy compounds such as perfluorooctane sulfonate (PFOS), perfluorooctanoic acid (PFOA) and perfluorohexane sulfonate (PFHxS) [7, 8]. Immune suppression, particularly reduced antibody responses to routine childhood vaccinations, is the most sensitive adverse outcome associated with PFAS exposure. On this basis, the European Food Safety Authority (EFSA) established a tolerable weekly intake (TWI) of 4.4 ng/kg body weight per week for the sum of PFOS, PFOA, PFHxS and PFNA [9]. Additional associations have been reported for dyslipidemia, liver enzyme alterations, reduced birthweight, and increased risks of kidney and testicular cancers, particularly at higher exposure levels [10–12]. Biomonitoring data confirm the widespread nature of exposure, with more than 96% of U.S. adolescents and adults showing detectable PFAS levels in 2017–2020, and with maternal–fetal transfer and breastfeeding contributing to PFAS body burdens early in life [13, 14].

Infants and young children are considered a particularly vulnerable population with respect to PFAS exposure due to a combination of physiological and behavioral factors. Compared with adults, infants experience higher intake per unit body weight, immature renal elimination pathways, and greater distribution of chemicals to rapidly developing organs [15]. These physiological characteristics coincide with distinct exposure behaviors, including frequent hand-to-mouth activity, object mouthing, and elevated ingestion of indoor dust, all of which can substantially increase internal exposure during critical windows of immune, endocrine and neurological development [16]. Recent studies suggest that textiles and consumer products intended for children may represent a non-negligible source of PFAS exposure. PFAS have been detected in 97% of tested car seats [17], in 87.9% of household textiles [18], and in childcare-environment dust [19]. Fluorinated side-chain polymers commonly used to impart water- and stain-repellent properties can release short-chain PFAS and precursor compounds during normal use, with migration demonstrated into artificial sweat and other physiological simulants, particularly for substances containing eight or fewer fluorinated carbons [17]. Despite these observations, comparatively little attention has been given to articles specifically designed for infants, such as comforters and stuffed toys, which are subject to repeated and prolonged mouthing and therefore represent a plausible route for salivary mobilization of PFAS.

Assessing PFAS migration under realistic exposure conditions is therefore essential for refining early-life exposure estimates. As demonstrated in recent studies investigating the release of organic contaminants from children's products, saliva-based migration experiments represent a relevant and standardized approach to

simulate oral exposure associated with infant mouthing behavior, and have been successfully applied to a range of consumer articles [20–23]. Building on these established methodologies, the present study quantitatively investigates PFAS migration from commercially available stuffed toys into artificial saliva as an age-appropriate simulant of infant oral exposure. Migration experiments were combined with an analytical assessment based on Liquid Chromatography–Tandem Mass Spectrometry (LC–MS/MS), allowing detailed quantification of PFAS-specific release profiles [24–26]. The resulting migration data were subsequently integrated into a deterministic exposure assessment framework specifically tailored to infants and toddlers, including intake estimation and margin of exposure calculations [20, 27, 28].

To further contextualize the potential contribution of stuffed toys to overall PFAS exposure, a reverse dosimetry approach based on plasma biomonitoring data was applied, following established exposure assessment frameworks for PFAS [29–32]. Plasma concentrations were selected from the literature over urinary biomarkers due to the long biological half-lives, strong serum protein binding, and pronounced bioaccumulation potential of many PFAS [33–37]. In addition, the effect of a standardized pre-washing step, reflecting common consumer practices for infant textiles, was evaluated to assess its potential as a practical exposure-mitigation measure and to explore compound- and product-specific responses, acknowledging that washing effects on PFAS mobility in textiles are heterogeneous and remain insufficiently characterized.

## MATERIALS AND METHODS

### Chemicals and standards

LC–MS grade methanol (MeOH, Honeywell, Charlotte, NC, USA), acetonitrile (ACN, Honeywell, Charlotte, NC, USA), ultrapure water (Sartorius Arium® Pro, Göttingen, Germany), ammonium acetate (Thermo Scientific, Waltham, MA, USA), ammonia solution (Merck, Darmstadt, Germany), sodium chloride, magnesium sulfate, magnesium chloride, calcium chloride, dipotassium hydrogen phosphate, potassium carbonate, potassium chloride, sodium nitrite (all Sigma-Aldrich, St. Louis, MO, USA), formic acid (Carl Roth, Karlsruhe, Germany), acetic acid (Sigma-Aldrich, St. Louis, MO, USA), and dichloromethane (Carl Roth, Karlsruhe, Germany) were used throughout the analytical procedures. PFAS standards were obtained from Neochema (Quebec, Canada) and comprised 24 target compounds, including Capstone B (CSPB); perfluorobutanoic acid (PFBA); perfluoropentanoic acid (PFPeA); perfluorohexanoic acid (PFHxA); perfluoroheptanoic acid (PFHpA); perfluorooctanoic acid (PFOA); perfluorononanoic acid (PFNA); perfluorohexanesulfonic acid (PFHxS); perfluorobutanesulfonic acid (PFBS); perfluorooctanesulfonic acid (PFOS); perfluorodecanoic acid (PFDA); perfluoroundecanoic acid (PFUnA); perfluorododecanoic acid (PFDoDA); sodium perfluoropentanesulfonate (PFPeS); perfluorooctanesulfonamide (PFOSA); perfluorotridecanoic acid (PFTrDA); perfluoroheptanesulfonic acid (PFHpS); sodium perfluorononanesulfonate (PFNS); perfluorodecanesulfonic acid (PFDS); 7H-perfluoroheptanoic acid (7H-PFHpA); sodium perfluoroundecanesulfonate (PFUnS); 1H,1H,2H,2H-perfluorohexanesulfonic acid (4:2 FTS); 1H,1H,2H,2H-perfluorooctanesulfonic acid (6:2 FTS); 1H,1H,2H,2H-perfluorodecanesulfonic acid (8:2 FTS); and perfluoro-3,7-dimethyloctanoic acid (PFOA-dimethyl). Detailed information on the target PFAS is provided in Table S1. Isotopically labeled internal standards (ISTDs) were purchased from Wellington Laboratories (Guelph, Ontario, Canada) and comprised $^{13}C_2$-PFHxA, $^{13}C_8$-PFOA and $^{13}C_8$-PFOS, used for quantification and correction of matrix and extraction effects. Native PFAS and ISTD working solutions were prepared in MeOH at 1000, 100, 10 and 1 µg/l. All solutions were stored in amber glass vials ( − 18°C for stocks; 4°C for working solutions). All consumables in contact with samples, including polypropylene centrifuge tubes (Greiner Bio-One, Kremsmünster, Austria), syringes, nylon syringe filters (0.45 µm, BGB Analytik, Bochum, Germany) and dispersive Solid Phase Extraction tubes, were certified PFAS-free. PTFE-containing materials were strictly avoided.

### Stuffed toys

A total of 18 commercially available stuffed toys were selected to represent the diversity of products commonly marketed for infants and toddlers in Europe (Type 1 to Type 18, Table S2). The selection included a range of brands, designs and fabric types, with items typically intended for children aged 0–3 years. For each stuffed toy, a specific sampling area was selected based on its likelihood of being mouthed by young children. Sampling focused on protruding or easily accessible parts such as ears, wings, body extremities or corners, which are commonly chewed or sucked during use [38, 39]. These areas are also more likely to undergo repeated wetting by saliva, making them particularly relevant for evaluating potential oral exposure [38]. All stuffed toys were handled using PFAS-free gloves and tools to minimize the risk of external contamination. Prior to migration experiments, samples were stored individually in clean aluminium foil at room temperature and protected from contact with potential PFAS-containing materials. An overview of the analysed products, including visual appearance and the selected sampling areas, is provided in Table S2.

### Migration into artificial saliva

Artificial saliva was prepared according to the BS 6684 - British Standard Specification for Safety Harnesses (1987). Briefly, 0.163 g of magnesium chloride ($MgCl_2$), 0.147 g of calcium chloride ($CaCl_2$), 0.575 g of dipotassium hydrogen phosphate ($K_2HPO_4$), 0.525 g of potassium carbonate ($K_2CO_3$), 0.327 g of sodium chloride (NaCl), and 0.746 g of potassium chloride (KCl) were dissolved in 1 L of water. The pH was adjusted to 6.5 using glacial acetic acid [40]. Artificial saliva was freshly prepared and stored at 4 °C prior to use. Migration experiments were designed to simulate infant mouthing behavior. For each assay, a 10 $cm^2$ section of stuffed toy fabric was individually placed in a 40 ml borosilicate glass vial containing 10.0 ml of artificial saliva. Vials were sealed and incubated for 30 min at 37 °C under orbital agitation at 140 rpm, following established saliva-migration protocols for children's products [20, 41–44].

### PFAS migration and extraction following saliva exposure

Following saliva incubation (Sections 2.3), 8.0 ml of artificial saliva were transferred into a 50 ml polypropylene centrifuge tube. Isotopically labeled PFAS ISTD were added at a final concentration of 500 ng/l to each sample prior to extraction. PFAS were extracted using a QuEChERS-based protocol (Figure S1). QuEChERS extraction was performed according to previous studies [24, 45]. Briefly, 10.0 ml of ACN and 150 µL of formic acid were added to each tube, then 4.0 g of anhydrous magnesium sulfate ($MgSO_4$) and 1.0 g of sodium chloride (NaCl) were added to induce salting-out. Tubes were immediately shaken vigorously for 1 min and centrifuged for 10 min at 3000 rpm. After phase separation, 7.0 mL of the upper ACN layer were transferred into a 15 mL polypropylene tube containing dispersive solid-phase extraction (dSPE) sorbent (XX Mix Chromabond®, Macherey-Nagel, Germany). Samples were shaken for 1 min and centrifuged again for 10 min at 3000 rpm. The cleaned supernatant was filtered through PFAS-free 0.45 µm nylon syringe filters into clean 15 mL tubes. Extracts were evaporated to dryness overnight at 70°C, reconstituted in 1.0 mL of MeOH/water (50:50, v/v) containing 0.1% formic acid, filtered again through PFAS-free 0.45 µm nylon filters, and transferred into LC vials for LC–MS/MS analysis.

### Effect of pre-washing on PFAS migration

To assess the effect of washing as a potential exposure-mitigation measure, a second experimental condition was evaluated following the same migration and extraction protocol described above (Sections 2.3 and 2.3.2; Fig. S1), with the addition of a washing step prior to saliva incubation. For this condition, 10 $cm^2$ fabric sections were placed in 100 ml of Evian® bottled water and shaken for 1 h at 30°C under orbital agitation at 200 rpm to simulate the mechanical movements of a domestic washing-machine cycle. No detergent was used during this step to avoid surfactant-induced ion suppression. After washing, fabric sections were air-dried at room temperature for 45 min and subsequently incubated in artificial saliva and processed identically to the non-washed samples.

### PFAS analysis

PFAS analysis was performed by LC–MS/MS using an LCMS-8060NX instrument (Shimadzu) equipped with a Shim-pack GIST C18 column (4.6 × 150 mm, 5.0 µm, Shimadzu Scientific). Chromatographic separation was carried out at a column temperature of 45°C with an injection volume of 20 µL. A binary gradient was applied at a constant flow rate of 0.50 mL/min using mobile phase A consisting of a 2 mM ammonium acetate solution prepared in an ACN/water mixture, and mobile phase B consisting of pure ACN. The ammonium acetate buffer was prepared by dissolving 0.38 g of

ammonium acetate in 1 L of ultrapure water, followed by sonication for 20 min, filtration through a 0.45 µm nylon membrane. The gradient program started at 80% A and 20% B, was linearly increased to 100% B by 10 min, held at 100% B until 13 min, and then returned to 80% A and 20% B at 13.01 min, followed by re-equilibration until the end of the run at 23 min. The mass spectrometer was operated in electrospray ionization in negative polarity mode (ESI–). The nebulizing gas flow was set at 3 l/min, while both drying and heating gas flows were set at 10 l/min. The interface temperature was maintained at 250°C, the desolvation temperature at 444°C, the desolvation line (DL) temperature at 200°C, and the heat block temperature at 350°C. Quantification was performed in multiple reaction monitoring (MRM) mode. Compound-specific MRM transitions, collision energies and retention times are reported in Table S3.

## Quality assurance and quality control

Quality assurance and quality control procedures were applied throughout the analytical workflow. Quantification was based on calibration curves prepared in spiked artificial saliva over a concentration range of 1 to 3000 ng/l and corrected using ISTD. The artificial saliva matrix was prepared fresh and identically for every analytical batch, ensuring a consistent calibration matrix and limiting sample-to-sample variability in matrix effects. The three ISTDs were assigned to target PFAS on a structural class basis: $^{13}C_2$-PFHxA was used for short-chain perfluoroalkyl carboxylic acids (PFBA, PFPeA, PFHxA, PFHpA), $^{13}C_8$-PFOA for medium- and long-chain perfluoroalkyl carboxylic acids (PFOA, PFNA, PFDA, PFUnA, PFDoDA, PFTrDA, 7H-PFHpA, PFOA-dimethyl), and $^{13}C_8$-PFOS for perfluoroalkyl sulfonates and fluorotelomer-derived analytes (PFBS, PFPeS, PFHxS, PFHpS, PFOS, PFNS, PFDS, PFUnS, PFOSA, CSPB, 4:2 FTS, 6:2 FTS, 8:2 FTS). The corresponding ISTD assignment for each target PFAS is also reported in Table S1. Calibration linearity was considered acceptable when the coefficient of determination ($R^2$) exceeded 0.990. Analytical sensitivity was assessed using a signal-to-noise (S/N) approach, with compound-specific limits of quantification (LOQ) defined at S/N ratios of 10:1. Procedural blanks, recovery experiments and duplicate analyses were systematically included to monitor data quality. Duplicate analyses were performed at least once per analytical sequence and at a minimum frequency of one duplicate every 9 samples, with acceptable variability defined as a relative difference within ±20%. Recovery experiments were considered acceptable when recoveries ranged between 70% and 130% of the target value. Recoveries from spiked artificial saliva ($N = 4$) fell within this range for 23 of the 24 target PFAS, with only 4:2 FTS slightly exceeding the upper bound (168%). Variability across replicates was generally low (SD < 25%), supporting effective compensation of matrix-related signal suppression by the assigned ISTDs. LOQ, recoveries (mean ± SD) and $R^2$ values are reported in Table S4. Dedicated blanks of the Evian bottled water used for experiments were also analysed, and all target PFAS were below the respective LOQ.

## Total daily intake (TDI) calculation

To quantify the potential oral exposure to PFAS resulting from mouthing of stuffed toys, the total daily intake (TDI) was estimated using a deterministic exposure approach. The TDI represents the mass of contaminant potentially absorbed per unit of body weight and per day (ng/kg/day) and reflects internal exposure associated with saliva-mediated transfer from contaminated textiles. The TDI was calculated according to Eq. (1):

$$TDI(ng/day/kg) = \frac{M_f C_s V_s D_{abs}}{B_w} \quad (1)$$

where $M_f$ is the mean daily mouthing frequency, set at 0.77 h/day for infants according to the literature [46, 47], $C_s$ is the concentration of each PFAS measured in artificial saliva (ng/ml), $V_s$ is the estimated volume of saliva in contact with the textile, assumed to be 19 ml/h according to previous studies [20, 48, 49], $D_{abs}$ is the digestive absorption rate for PFAS settled at 100% following conservative assumptions commonly adopted in regulatory risk assessments [33], and $B_w$ is the average body weight, assumed to be 9.5 kg for a one-year-old child [46]. The corresponding weekly intake was derived by multiplying the TDI by seven days.

## Exposure assessment and margin of exposure approach

To characterize the potential health relevance of PFAS exposure resulting from mouthing of stuffed toys, a margin of exposure (MOE) approach was applied. This method allows comparison between estimated exposure levels and toxicological reference values and is commonly used for screening-level risk characterization of chemical mixtures in consumer exposure scenarios [20, 50–52]. The MOE was calculated for each individual PFAS as the ratio between a toxicological reference point and the estimated daily exposure (DE), as shown in Eq. (2):

$$MOE = \frac{NOAEL}{TDI} \quad (2)$$

where NOAEL represents the No Observed Adverse Effect Level (ng/kg body weight/day) for the individual PFAS, and TDI corresponds to the daily exposure dose estimated from saliva-mediated migration during mouthing (Section 2.8, Eq. 1). NOAEL values were derived from the most recent toxicological literature and refer to developmental effects, which represent the most sensitive endpoints for PFAS [32]. Specifically, NOAEL values of 515 ng/kg/day for PFOS and 1000 ng/kg/day for PFNA were used [32]. These values were applied as reference points for all other PFAS belonging to the same chemical class, i.e. sulfonates and carboxylic acids, respectively. To account for combined exposure to multiple PFAS migrating from the same stuffed toy, a total margin of exposure ($MOE_{total}$) was calculated using a dose-addition approach, as expressed in Eq. (3):

$$MOE_{total} = \frac{1}{\sum_{i=1}^{n}\left(\frac{1}{MOE_i}\right)} \quad (3)$$

where $MOE_i$ represents the margin of exposure of each individual PFAS detected in artificial saliva. This approach assumes additivity of effects and is considered appropriate for compounds with similar modes of action or overlapping toxicological endpoints [20, 50, 51]. An MOE or $MOE_{total}$ value > 100 was interpreted as indicating no appreciable health concern, whereas values below this threshold suggest a reduced margin between exposure and effect levels and warrant closer consideration [52, 53].

## Contribution of stuffed toys to total PFAS exposure

To estimate the contribution of PFAS exposure from stuffed toys to overall PFAS exposure, a reverse dosimetry approach based on measured plasma concentrations was applied. All calculations were performed using plasma concentrations and assume linear toxicokinetics and steady-state conditions [54, 55]. For each PFAS, an equivalent total daily intake was derived from plasma concentrations according to Eq. (4):

$$Intake_{plasma,i}(ng/kg/day) = \frac{C_{plasma,i}(ng/ml)CL_i(ml/kg/day)}{D_{abs}} \quad (4)$$

Where $C_{plasma,i}$ is the PFAS concentration measured in plasma, $CL_i$ is the body-weight–normalized clearance and $D_{abs}$ is the digestive absorption rate for PFAS as described in Section 2.8. The relative contribution of PFAS exposure from stuffed toys was then expressed as a percentage of the total intake estimated from plasma concentrations using Eq. (5):

$$\%Contribution_i = 100 \times \frac{TDI_{stuffed\ toy,i}}{Intake_{plasma,i}} \quad (5)$$

Where $TDI_{stuffed\ toy,i}$ represents the daily intake estimated from saliva-mediated PFAS migration during mouthing, calculated through Eq. (1).

Individual reverse dosimetry calculations were performed for PFAS for which compound-specific plasma concentrations and toxicokinetic parameters were available from the literature, namely PFOS, PFHxS, PFOA and PFNA. For PFHpS, PFBS, PFDA, PFUnDA, PFUdA and PFHpA [31, 32, 56], plasma concentrations were available but compound-specific clearance values were not consistently reported (Table S5). In these cases, clearance was approximated using the mean clearance of structurally related reference compounds, calculated separately for sulfonates (mean of PFOS and PFHxS) and carboxylic acids (mean of PFOA and PFNA)(Table S5). For all remaining PFAS included in the study for which individual plasma concentrations were not available, a class-based approach was applied in which the combined contribution of multiple PFAS was evaluated rather than that of individual compounds [54, 55]. In this case, representative plasma concentrations for perfluoroalkyl carboxylic acids were estimated using the mean plasma concentration of PFNA, PFDA, PFUnDA, PFUdA and PFHpA, while those for perfluoroalkyl sulfonates were estimated using the mean plasma concentration of PFHxS, PFHpS and PFBS (Table S5). These class-specific concentrations were used to estimate intake and contribution for the sum of carboxylates and the sum of sulfonates, respectively. PFOS and PFOA were excluded from these class-based calculations because their plasma concentrations were markedly higher than those of the other PFAS and would otherwise disproportionately influence the estimates, whereas the remaining compounds showed more comparable concentration ranges (Table S5). Clearance values for these grouped

**Table 1.** Detection frequency (DF, %), mean and range of PFAS migration levels (µg/m²), and mean and range of percent composition relative to ΣPFAS for the 24 target PFAS analyzed in stuffed toy samples.

| Compound | DF (%) | Mean [µg/m²] | Range [µg/m²] | LOQ [µg/m²] | Mean composition (%) | Range composition (%) |
|---|---|---|---|---|---|---|
| CSPB | 0 | <LOQ | <LOQ | 0.50 | — | — |
| PFBA | 100 | 4.00 | 0.58 – 30.61 | 0.50 | 20.1 | 2.2 – 90.6 |
| PFPeA | 94 | 0.88 | <LOQ – 2.62 | 0.25 | 10.2 | <LOQ – 30.9 |
| PFHxA | 100 | 21.97 | 0.22 – 120.73 | 0.10 | 38.7 | 0.9 – 92.4 |
| PFHpA | 100 | 0.19 | 0.12 – 0.28 | 0.10 | 2.4 | 0.1 – 6.1 |
| PFOA | 94 | 0.20 | <LOQ – 0.43 | 0.01 | 2.6 | <LOQ – 10.6 |
| PFNA | 83 | 0.03 | <LOQ – 0.11 | 0.01 | 0.4 | <LOQ – 1.4 |
| PFHxS | 6 | 0.28 | <LOQ – 0.28 | 0.25 | 0.4 | <LOQ – 6.4 |
| PFBS | 33 | 0.11 | <LOQ – 0.41 | 0.25 | 1.7 | <LOQ – 8.2 |
| PFOS | 100 | 0.25 | 0.07 – 0.79 | 0.05 | 2.1 | 0.1 – 5.5 |
| PFDA | 89 | 0.07 | <LOQ – 0.10 | 0.05 | 0.8 | <LOQ – 2.3 |
| PFUnA | 0 | <LOQ | <LOQ | 0.10 | — | — |
| PFDoDA | 61 | 0.02 | <LOQ – 0.05 | 0.01 | 0.2 | <LOQ – 0.6 |
| PFPeS | 89 | 0.18 | <LOQ – 0.29 | 0.05 | 1.6 | <LOQ – 5.3 |
| PFOSA | 0 | <LOQ | <LOQ | 1.00 | — | — |
| PFHpS | 100 | 2.15 | 0.10 – 7.57 | 0.10 | 14.8 | 0.3 – 45.4 |
| PFNS | 0 | <LOQ | <LOQ | 0.05 | — | — |
| PFDS | 0 | <LOQ | <LOQ | 0.05 | — | — |
| 7H-PFHpA | 89 | 0.19 | <LOQ – 0.39 | 0.10 | 2.3 | <LOQ – 6.6 |
| PFOA-dimethyl | 0 | <LOQ | <LOQ | 0.25 | — | — |
| PFUnS | 0 | <LOQ | <LOQ | 0.50 | — | — |
| 4:2 FTS | 0 | <LOQ | <LOQ | 0.10 | — | — |
| 6:2 FTS | 89 | 0.21 | <LOQ – 1.31 | 0.05 | 1.9 | <LOQ – 6.5 |
| 8:2 FTS | 0 | <LOQ | <LOQ | 0.01 | — | — |

Limits of quantification (LOQ) are reported in µg/m² for direct comparison with measured values. Values reported as <LOQ refer to concentrations below the corresponding analytical limit of quantification. Percent composition values are calculated for each toy as the contribution of each PFAS to ΣPFAS in that sample, and summarized as the mean and range across the 18 stuffed toys.

calculations were assigned using the same class-based approach described above.

## RESULTS

### Frequency of detection of PFAS

Among the targeted PFAS, a subset of compounds was not detected ( < LOQ) in any of the analysed stuffed toy samples, including CSPB, PFUnA, PFOSA, PFTrDA, PFNS, PFDS, PFOA-dimethyl, PFUnS, 4:2 FTS and 8:2 FTS (Table 1). By contrast, a broad range of PFAS was detected across the dataset, with multiple compounds quantified in the majority of samples. In particular, PFBA, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFOS, PFDA, PFPeS, PFHpS, 7H-PFHpA and 6:2 FTS were detected in more than 80% of the analysed stuffed toys (Table 1). Within this group, short-chain perfluoroalkyl carboxylic acids (PFCAs), particularly PFBA, PFPeA and PFHxA, showed the highest overall occurrence. PFHxA was detected in all samples and exhibited the widest concentration range. PFHpS was also detected in all samples and consistently occurred at elevated levels relative to other sulfonates. A second group of PFAS showed similarly high detection frequencies but occurred at clearly lower concentrations compared to the dominant contributors. This group included PFHpA, PFOA, PFPeS, 7H-PFHpA and 6:2 FTS (Table 1). These compounds were present in most toys but contributed less to the overall PFAS burden. Other PFAS, such as PFNA and PFDA, were frequently detected but at comparatively low concentrations, resulting in a minor contribution to total PFAS levels despite their high occurrence. Finally, PFBS and PFDoDA displayed low detection frequencies but reached relatively elevated concentrations when present.

### Total PFAS release across stuffed toys

The compound-resolved release profiles indicate that PFAS were detected in all analyzed stuffed toys, although with pronounced differences in both concentration levels and compound-specific patterns across product types (Fig. 1). Most samples show low to moderate concentrations across the majority of PFAS, whereas a limited number of toys display markedly elevated levels driven by a single dominant compound. In particular, stuffed toys Type 2, Type 10, Type 12 and Type 13 exhibit substantially higher PFAS concentrations compared to the rest of the dataset. As shown in Fig. 1, these samples are characterized by PFHxA, which largely determine their overall PFAS profiles. In contrast, toys with low total PFAS release generally show consistently low PFHxA levels and relatively uniform concentrations across the remaining compounds. Across all samples, PFHxA shows the greatest inter-sample variability, spanning more than two orders of magnitude and clearly distinguishing high-release from low-release toys. Other PFAS contribute to the overall profiles but remain comparatively stable across toy types and do not drive the observed extremes. When concentrations are summed across compounds, the same subset of toys with elevated PFHxA concentrations also displays the highest ΣPFAS values (Fig. 1), confirming that differences between products are primarily associated with compound-specific concentration patterns rather than uniform shifts across all PFAS. In quantitative terms, PFHxA emerged as the dominant compound in the migration profile,

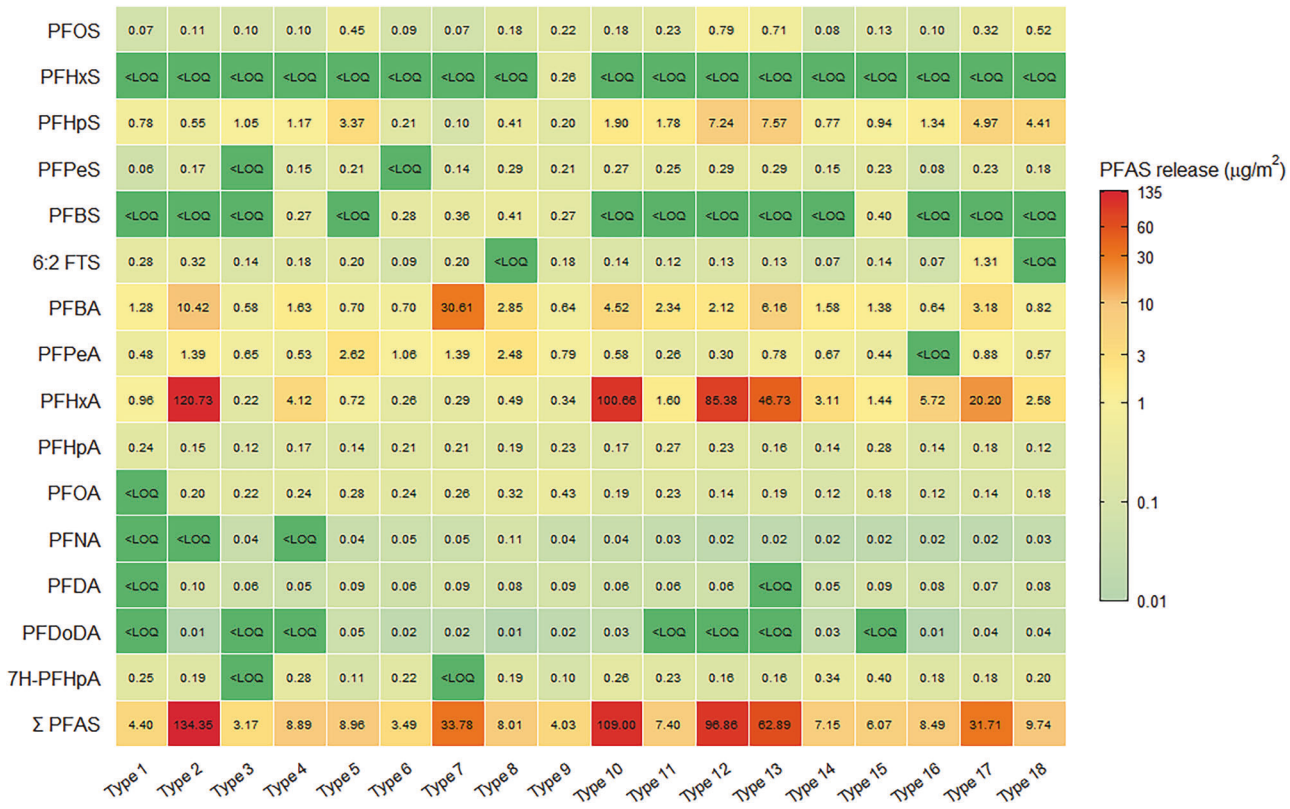



**Fig. 1 Heatmap of PFAS concentrations released from stuffed toys into artificial saliva.** Rows represent individual PFAS, columns correspond to the 18 stuffed toy types analysed. Concentrations are expressed in µg/m² of fabric surface. Values below the limit of quantification are indicated as <LOQ. The bottom row reports the summed PFAS concentration (ΣPFAS) for each toy. Color intensity reflects relative concentration levels, from <LOQ (green) to high (red).

followed by PFBA, PFHpS and PFPeA (Table 1). All other detected PFAS contributed less than 3% on average to the total migration profile.

### PFAS weekly intake estimation

Estimated weekly PFAS intake via saliva-mediated mouthing shows compound-specific and product-specific variability (Fig. 2; Table S6). Several PFAS contribute measurably to the estimated intake in individual toy types, with weekly intake values reaching the order of tenths to units of ng/kg body weight/week for selected compounds and samples. Among individual PFAS, PFHxA represents the largest contribution to weekly intake in toys exhibiting elevated overall exposure, in line with previous results (Section 3.2). PFHpS also contributes appreciably to weekly intake estimates in a subset of toys, reaching values close to or above 1.0 ng/kg body weight/week in some cases. Other PFAS, including PFBA and PFPeA, contribute more uniformly across toy types but generally at lower levels. In contrast, long-chain PFCAs such as PFNA, PFDA and PFDoDA contribute marginally to the estimated intake despite being frequently detected, due to their low concentrations. The summed PFAS intake (ΣPFAS) exceeds 10.0 ng/kg body weight/week for a limited number of toy types (Fig. 2; Table S6). The EFSA has established a TWI of 4.4 ng/kg body weight/week for the combined exposure to PFOS, PFOA, PFHxS and PFNA. When the exposure assessment is restricted to these four compounds, estimated weekly intakes range from 0.1 to 1.0 ng/kg body weight/week depending on toy type, and no exceedance of the TWI is observed (Fig. 2; Table S6).

### Margin of exposure associated with PFAS exposure

The MOE calculated for individual PFAS spans several orders of magnitude across the analysed stuffed toys (Fig. 3; Table S7). For most compounds and toy types, MOE values are well above the reference threshold of 100, indicating a large margin between estimated exposure and toxicological reference points. However, a limited number of individual PFAS exhibit MOE values below or close to 100, indicating reduced margins of exposure under conservative assumptions [20, 52, 53]. These low MOE values are observed almost exclusively for short-chain perfluoroalkyl carboxylic acids, in particular PFHxA, which shows MOE values < 100 for several toy types (notably Types 2, 10, 12, 13 and 17), reflecting its relatively high migration into artificial saliva combined with the class-based toxicological reference value applied. When considering combined exposure, the MOEtot varies markedly between toy types (Fig. 3; Table S6). MOEtot values fall <100 for a limited number of samples (Types 2, 10, 12 and 13), while remaining above 100 for all other toy types. These lowest MOEtot values coincide with samples characterized by elevated migration of PFHxA and, to a lesser extent, PFHpS, indicating that the cumulative risk metric is driven by a small subset of highly migrating compounds rather than by uniform contributions across all PFAS.

### Contribution of stuffed toys to total daily PFAS intake

The relative contribution of PFAS exposure from stuffed toys to the TDI estimated from plasma concentrations (Table S8) shows a wide variability across compounds and toy types (Fig. 4). For several PFAS, the estimated daily intake associated with saliva-mediated mouthing represents a non-negligible fraction of the total intake inferred from plasma levels, and in many cases exceeds 100%, indicating that the exposure derived from stuffed toys alone is comparable to or higher than the intake back-calculated from biomonitoring data. The highest relative contributions are consistently observed for perfluoroalkyl sulfonates,

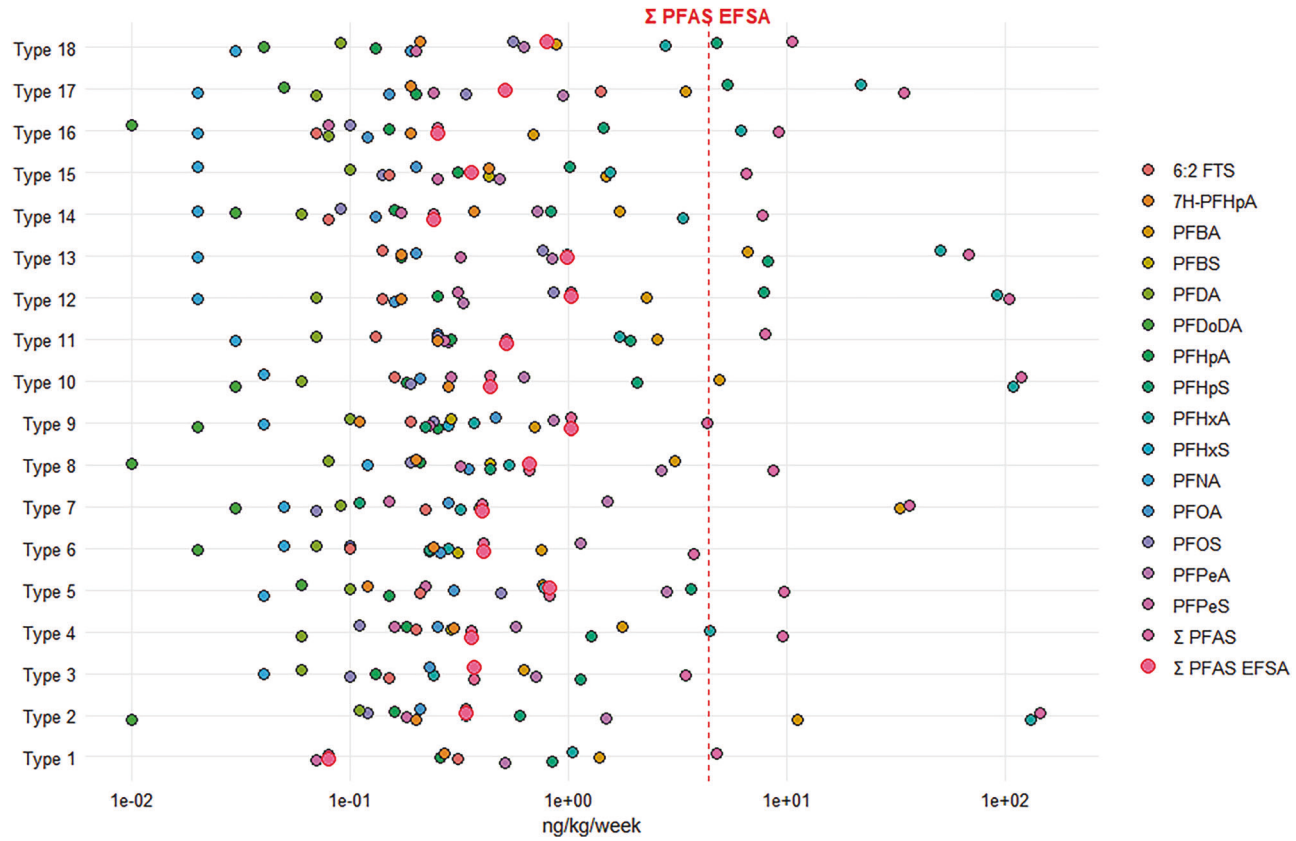



**Fig. 2 Estimated weekly intake of PFAS via saliva-mediated mouthing of stuffed toys, expressed as ng/kg body weight/week.** Each point represents the estimated weekly intake for an individual PFAS and toy type. The summed intake for EFSA-regulated PFAS (sum of PFOS, PFOA, PFHxS and PFNA; Σ PFAS EFSA) is shown and highlighted with red-circled markers. The red dashed vertical line marks the EFSA tolerable weekly intake (TWI) of 4.4 ng/kg body weight/week for Σ PFAS EFSA, and is labeled accordingly. Values are displayed on a logarithmic scale.

particularly PFHpS and PFBS. For these compounds, contributions frequently exceed several hundred % and reach very high values in multiple toy types, reflecting both elevated migration into saliva and low plasma-based intake estimates derived from clearance assumptions (Table S8). PFPeS also shows systematically high contributions, generally above 150–300% across most samples. Among perfluoroalkyl carboxylic acids, medium- and long-chain compounds display contrasting behaviors. PFHxA exhibits moderate to high relative contributions across several toy types, with values exceeding 100% in selected samples, underscoring its disproportionate role in saliva-mediated exposure compared to other carboxylates. PFDA and PFDoDA frequently exhibit contributions well above 100%, in some cases exceeding several thousand %, highlighting a substantial relative contribution of stuffed toys to overall exposure for these longer-chain carboxylates. In contrast, PFHpA generally shows intermediate contributions, often close to or slightly above 100%, whereas PFNA and PFOA consistently contribute less than 100% across all samples. For short-chain carboxylic acids such as PFBA and PFPeA, the contribution of stuffed toys to total intake is negligible or close to zero in most cases, reflecting low saliva-mediated intake relative to plasma-based exposure estimates.

### Effect of washing on PFAS migration and exposure

The % reduction after washing indicates an overall decrease in PFAS migration for many compounds and toy types, with numerous cases showing complete removal (100%), consistent with a decrease to <LOQ (Fig. 5). However, the response to washing is heterogeneous across both compounds and products. For the dominant contributors to the migration profiles, washing is generally effective. PFHxA shows high and often complete reductions (95–100% for most toy types), while PFOS is consistently reduced by 100% across all samples. PFHpA also exhibits complete reduction in the majority of toys, with only a few cases showing partial reductions (e.g. Type 11 and Type 12). In contrast, several PFAS display no measurable effect of washing (0%) or even increased migration after washing, reflected by negative reduction values (Fig. 5). This behavior is particularly evident for PFBA and PFPeA in several toy types, where strongly negative values indicate a post-washing release. A similar pattern is observed for ΣPFAS, which shows positive reductions in most samples but also clear increases in a subset of toy types, including pronounced increases for Type 11 and Type 15. At the MOE level, no values fall into a range of concern after washing (MOE < 100), indicating that the MOE remains favorable even in toy types showing increased post-washing migration for individual PFAS (Table S9).

## DISCUSSION

### Ubiquitous PFAS release from stuffed toys under mouthing-relevant conditions

Numerous studies demonstrate that PFAS are widespread across consumer products, with textiles consistently among the most affected categories. Large-scale screenings have reported PFAS in the majority of tested consumer articles, with textiles showing both high detection frequencies and the greatest chemical diversity [57]. Targeted surveys similarly detected PFAS in up to 87.9% of household and children's textiles, indicating that their presence is ubiquitous rather than incidental [18]. Soft consumer products have been identified as relevant contributors to early-life PFAS exposure. In childcare environments, Zheng et al. (2019)

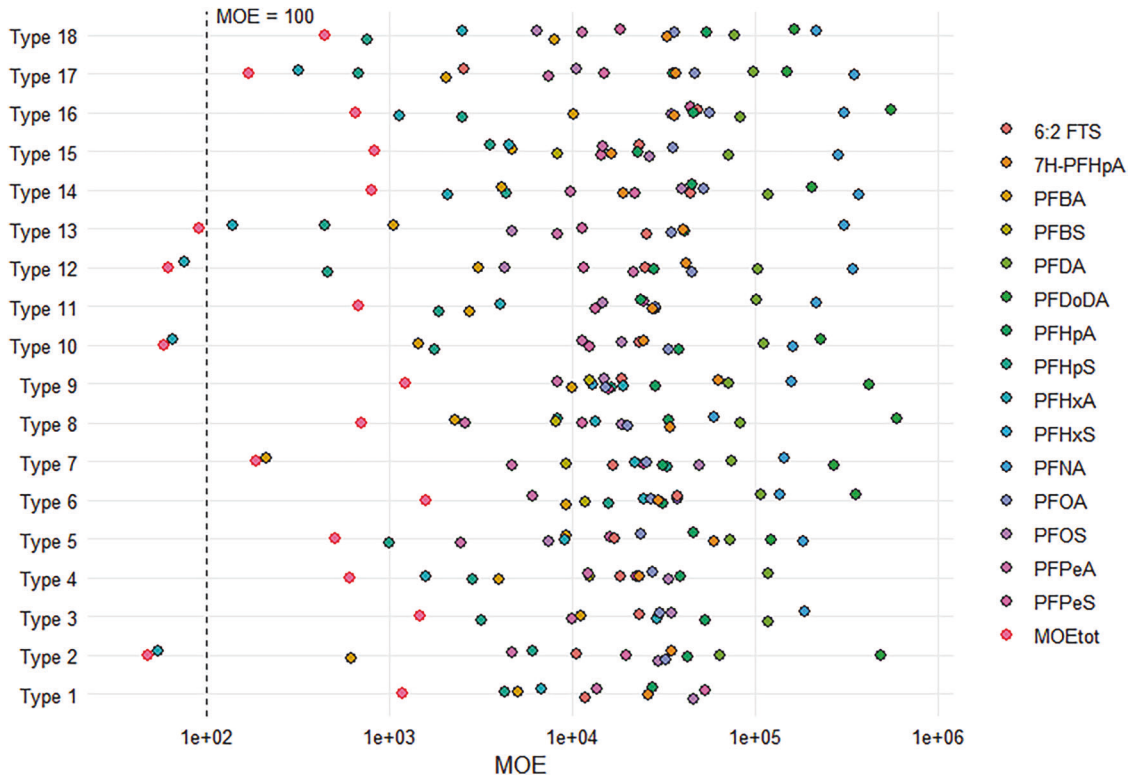



**Fig. 3 MOE for individual PFAS and combined exposure (MOEtot) associated with saliva-mediated mouthing of stuffed toys.** Each point represents the MOE calculated for a specific PFAS and toy type, while MOEtot reflects the combined exposure across compounds. The dashed vertical line indicates the reference MOE value of 100. Values are displayed on a logarithmic scale.

detected 28 PFAS in dust and nap mats at concentrations ranging from 8.1 to 3700 ng/g, highlighting the importance of indoor materials [19]. Mouthing behavior represents a critical exposure pathway, as demonstrated by Aurisano et al. (2021), who reported saliva migration rates sufficient to contribute meaningfully to daily intake in infants and toddlers [58]. Together, these findings support the plausibility of saliva-mediated PFAS exposure from soft consumer goods.

Against this background, the present study shows that PFAS migrate from stuffed toys into artificial saliva under conditions approximating oral physiology. Notably, PFAS release was observed for all analysed toys, indicating the absence of non-emitting samples. Although migration levels varied markedly between products, this observation aligns with previous reports of widespread PFAS detection in textiles [57, 59]. Migration profiles were dominated by short-chain perfluoroalkyl carboxylic acids (PFCAs), particularly PFBA, PFHxA and PFPeA. Average concentrations in artificial saliva were highest for PFBA and PFHxA (approximately 4.0 and 22 μg/m$^2$, respectively), whereas most other PFAS occurred at substantially lower levels or were not detected.

This pattern is consistent with previous studies on PFAS-treated textiles, which reported preferential enrichment of short-chain PFCAs in aqueous media as a result of their higher mobility and lower sorption affinity compared to long-chain analogues [18, 59]. At the product level, variability in total PFAS release was driven by a limited number of toys exhibiting elevated ΣPFAS values (62.9–134.3 μg/m$^2$), almost entirely attributable to PFHxA. Similar behavior has been observed for other textile products, where exposure estimates are frequently dominated by one or a few highly mobile PFAS rather than by uniformly elevated concentrations across the full compound spectrum [60, 61].

These findings highlight PFHxA as a key determinant of saliva-mediated PFAS exposure from stuffed toys. The prominence of PFHxA is consistent with its increasing use as a short-chain replacement for regulated legacy PFAS such as PFOA and PFOS [62]. Short-chain PFAS, including PFHxA, have been shown to migrate more readily than long-chain PFAS across a range of consumer products and simulants, with enhanced migration under aqueous, acidic or alcohol-containing conditions [62]. Experimental studies on food contact materials further demonstrated that short-chain PFCAs are often nearly completely leachable in polar solvents and simulants, underscoring their high transfer potential relative to longer-chain PFAS [62]. Although migration data for plush textiles remain scarce, the present results extend these observations to stuffed toys, indicating that PFHxA exhibits efficient release under mouthing-relevant conditions. PFAS migration levels observed in this study fall within the low-to-medium range reported for textile products. While extractable PFAS concentrations in technical textiles or children's clothing can reach tens to hundreds of μg/m$^2$, saliva-mediated migration is typically reported in the low microgram per square meter range [59, 61, 63]. The present findings are therefore consistent with previous estimates suggesting that only a small fraction of textile-bound PFAS migrates into saliva. However, this fraction remains exposure-relevant given the frequency and duration of mouthing behavior during early life. Indeed, the consistent release of PFHxA and other short-chain PFCAs, combined with increasing detection of PFHxA in human biomonitoring matrices such as breast milk and urine [62], underscores the relevance of stuffed toys as a non-negligible source of oral PFAS exposure in early life, despite moderate migration levels relative to other textile categories.

### Interpretation of exposure metrics and contextualization of PFAS exposure from stuffed toys

To contextualize the potential health relevance of PFAS exposure resulting from mouthing of stuffed toys, this study combined a MOE approach with reverse dosimetry based on biomonitoring data. These complementary methods are widely applied in exposure

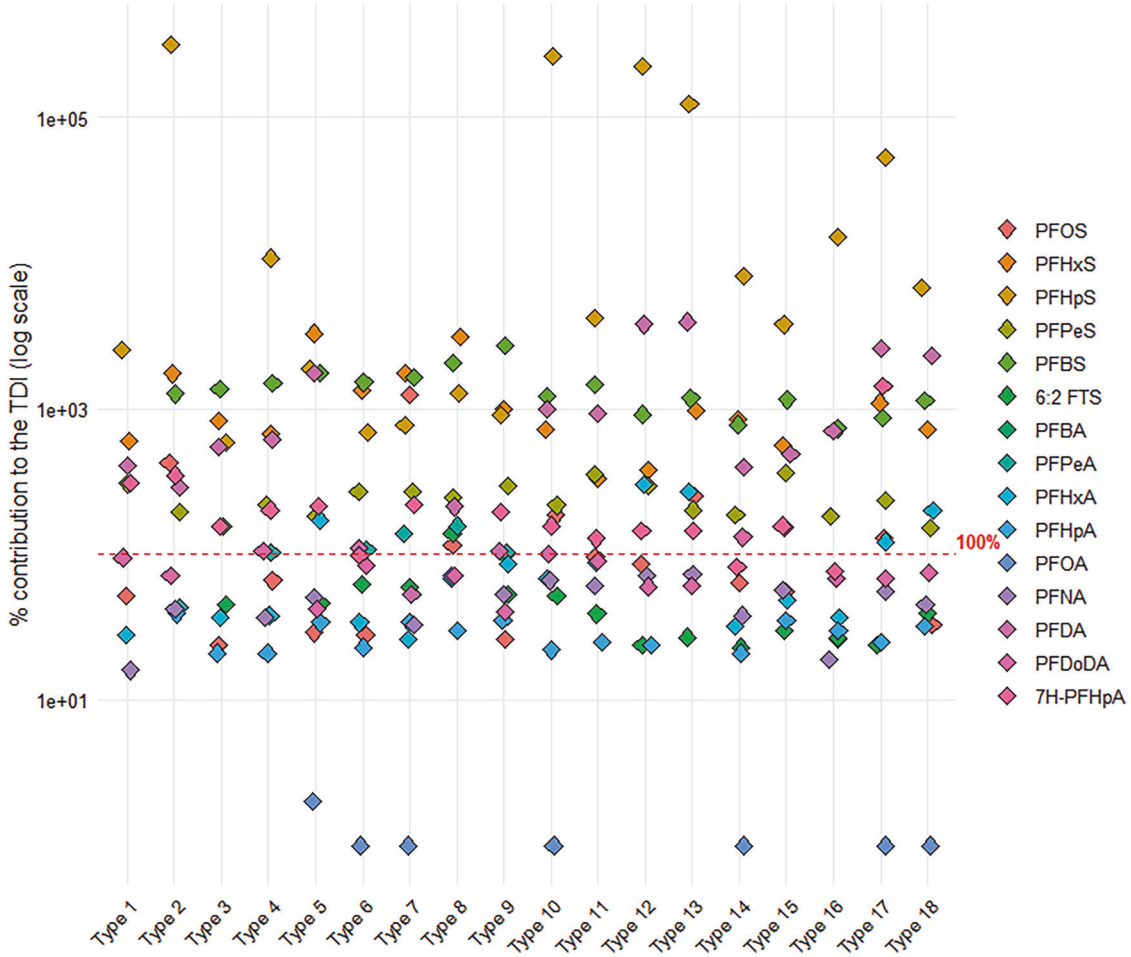



**Fig. 4 Relative contribution of PFAS exposure from stuffed toys to the tolerable daily intake (TDI), estimated by reverse dosimetry.** Values are expressed as the percentage of saliva-mediated intake during mouthing relative to the total PFAS intake inferred from plasma concentrations. Each point represents an individual PFAS for a given toy type. The dashed horizontal line at 100% indicates that the estimated intake from stuffed toys alone equals the intake inferred from plasma concentrations; values above 100% indicate a higher contribution from stuffed toys, while values below 100% indicate a lower contribution. Values are displayed on a logarithmic scale.

science as screening-level tools to prioritize exposure pathways and compounds rather than to quantify actual internal dose or health risk [20, 64–66]. In the present study, MOEs were calculated for individual PFAS using developmental toxicity reference values and combined into a total MOE assuming dose additivity, an approach commonly adopted for PFAS mixtures given their overlapping modes of action and shared target systems [9, 67]. Importantly, MOE-based metrics are not intended to represent precise estimates of internal exposure, but rather to identify exposure scenarios in which margins between exposure and effect levels may be reduced and warrant closer scrutiny. Across the analyzed compounds, clear differences in MOE patterns were observed. MOE values close to the reference threshold of 100 occurred for PFHpS in selected samples, whereas sulfonates with shorter chain length, such as PFBS, generally exhibited higher MOE values when detected. In contrast, PFOS consistently showed MOE values well above 100 across all toy types. Similarly, PFNA and longer-chain perfluoroalkyl carboxylic acids (PFDA, PFDoDA) displayed very high MOE values in all cases, often exceeding $10^4$–$10^5$, reflecting low estimated intakes combined with higher NOAEL reference values [32]. These patterns indicate that reduced margins of exposure were driven by a limited subset of compounds rather than by uniform contributions across the PFAS mixture.

In parallel, reverse dosimetry was applied following established literature approaches to contextualize PFAS exposure from stuffed toys relative to plasma-based intake estimates [54, 55, 68]. In several cases, the estimated contribution of PFAS exposure from stuffed toys exceeded 100% of the intake inferred from plasma concentrations. Such exceedances should not be interpreted as representing actual or cumulative human exposure. Rather, they reflect the conservative nature of the exposure assumptions applied, including upper-bound mouthing behavior, steady-state conditions, and the use of class-based toxicokinetic parameters where compound-specific data were unavailable. Previous studies have consistently shown that external dose-based exposure estimates can exceed biomonitoring-derived intakes by several-fold, particularly when single exposure pathways are assessed in isolation [68, 69]. Accordingly, these estimates should be interpreted as relative indicators of exposure relevance rather than absolute intake values. The combined use of MOE and reverse dosimetry highlights the complementary insights provided by different exposure metrics. While evaluation against health-based guidance values indicates that overall PFAS exposure from stuffed toys remains below the current EFSA TWI for regulated PFAS, the MOE-based analysis reveals reduced margins for specific compounds and products, particularly for short-chain PFAS such as PFHxA. This apparent divergence reflects the different purposes of these metrics: TWI values provide population-level benchmarks for cumulative exposure, whereas MOE and contribution analyses are more sensitive to pathway-

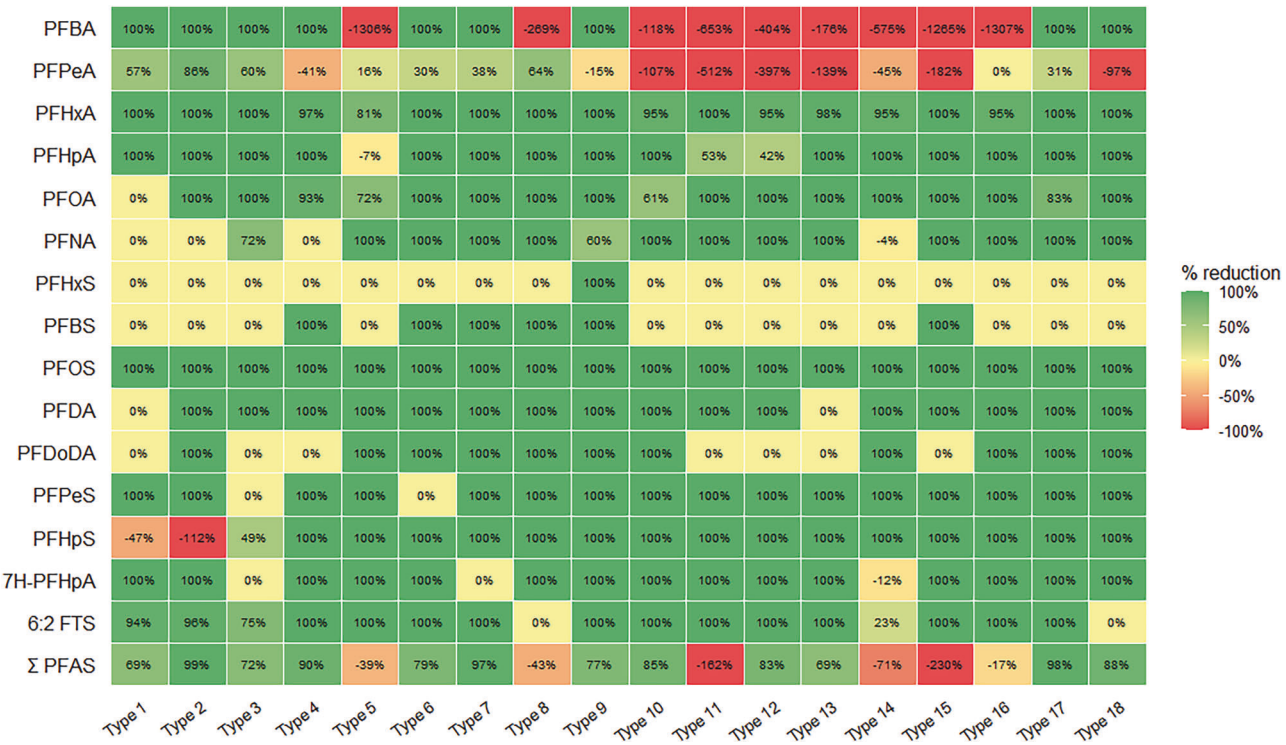



**Fig. 5 % change in PFAS migration after washing relative to the unwashed samples (% reduction) for each toy type.** Values of 0% indicate no difference; values > 0% indicate a reduction in migration (with 100% corresponding to complete removal to <LOQ); values < 0% indicate increased migration after washing. A diverging color scale is used with 0% as the central reference value.

specific and compound-specific drivers of exposure, especially under early-life scenarios.

Taken together, these findings illustrate a broader challenge in PFAS risk assessment, namely the potential mismatch between existing regulatory benchmarks and emerging exposure patterns driven by replacement PFAS. While compliance with current health-based guidance values such as the EFSA TWI suggests an overall low regulatory concern for legacy PFAS, the MOE- and contribution-based analyses highlight exposure relevance for short-chain and emerging PFAS, including PFHxA, for which toxicological and toxicokinetic data remain limited. This situation reflects a recurring issue in chemical substitution, where regulated substances are replaced by structurally related alternatives that are less well characterized from a toxicological perspective [70–72]. The present study underscores that such emerging PFAS can contribute meaningfully to exposure under realistic use conditions, even in products intended for sensitive populations. Moreover, these results indicate that consumer articles designed for infants and young children - particularly textile-based objects subject to frequent mouthing - remain insufficiently considered in current exposure assessment frameworks. Oral exposure pathways from textiles are still poorly described compared with dietary intake or dust ingestion, despite their relevance during early life. By combining saliva-mediated migration experiments with screening-level exposure metrics, this study provides early empirical evidence for an understudied exposure route and highlights the need for more comprehensive toxicological data, refined exposure models, and targeted regulatory attention for emerging PFAS in products intended for vulnerable populations.

### Effect of washing: mitigation potential and mechanistic considerations

Rinsing or washing is often recommended as a simple mitigation measure to reduce chemical exposure from consumer textiles. In the present study, most stuffed toys exhibited a decrease in ΣPFAS following rinsing, whereas a subset of products showed pronounced increases ( + 17 to +230%), driven almost entirely by PFBA. This heterogeneous response is consistent with the mixed evidence reported in the literature and indicates that washing does not produce a systematic or universally protective effect on PFAS migration. Indeed, previous studies have shown that washing can lead to both decreases and increases in extractable PFAS concentrations, depending on compound properties, fabric type, and treatment history [18, 73–75]. For textiles treated with durable water-repellent coatings, washing generally reduces perfluoroalkyl acid (PFAA) concentrations through leaching, but can simultaneously increase the extractable fraction of volatile or short-chain PFAS, particularly after aging [73]. Similarly, Schellenberger et al. (2019) demonstrated that washing releases PFAS-containing microfibers, with polyamide fabrics emitting substantially more fibers than polyester-based materials, thereby potentially increasing PFAS mobility and environmental dispersion [74]. Surveys of children's and household textiles further suggest that washing may redistribute PFAS within the textile matrix and increase surface accessibility [18]. Several mechanistic hypotheses may explain the bidirectional effects observed in this study. Washing may remove loosely bound PFAS through aqueous leaching, resulting in decreased migration for some compounds. Conversely, mechanical agitation and fiber swelling during washing may promote surface reorganization, abrasion of fluorinated coatings, or the release of PFAS previously trapped within the textile matrix, thereby increasing the accessibility and migration of highly mobile short-chain PFAS such as PFBA [18, 73, 74]. In addition, hydrolysis or transformation of residual precursors and side-chain fluorinated polymers during washing has been shown to increase the extractable concentrations of specific PFAS, particularly in fluorotelomer-based coatings, with effects that may be amplified by textile aging [59, 73]. In this context, 6:2 FTS, detected in the majority of stuffed toys analysed in this study, is of particular concern. As a fluorotelomer-based pre-PFAA structure, it can

undergo transformation under environmentally relevant conditions such as microbial activity and oxidative degradation into short-chain perfluorocarboxylic acids (PFCAs), including PFBA, PFPeA, and PFHxA [76, 77]. PFHxA is recognized as a major terminal degradation product of 6:2 fluorotelomer, with a toxicological profile that has been the subject of dedicated regulatory evaluation [78]. While the present study cannot directly attribute the post-washing increase observed for short-chain PFCAs to precursor transformation, the simultaneous presence of 6:2 FTS and short-chain PFCAs in the analysed products suggests that pre-PFAA precursors may contribute to the overall short-chain PFAS burden associated with stuffed toys and represent a relevant aspect for future investigations of textile-based exposure pathways.

### Relevance for early-life exposure assessment

Early life represents a critical window of susceptibility to PFAS exposure due to a combination of physiological vulnerability and distinct behavioral patterns, including higher intake per unit body weight and immature elimination pathways [9, 15, 79, 80]. Mouthing behavior, which is frequent and prolonged during infancy and early childhood, creates exposure scenarios that are not adequately captured by conventional dietary - or dust - based exposure models [6, 58, 81]. The present study directly addresses this gap by demonstrating saliva-mediated PFAS migration from stuffed toys under conditions approximating oral physiology, thereby providing empirical evidence for an exposure pathway that remains poorly represented in current assessment frameworks. Although stuffed toys are unlikely to constitute a dominant source of overall PFAS exposure, the consistent release observed across all tested products indicates that they represent a non-negligible contributor within the cumulative exposure profile of infants and toddlers. From an exposure science perspective, the approach adopted here is complementary to other screening tools such as the use of extractable organofluorine (EOF) analysis to identify materials or matrices associated with elevated PFAS exposure potential [82]. Saliva-mediated migration testing combined with screening-level exposure metrics provides a targeted approach to identify specific products, compounds and pathways that may disproportionately contribute to exposure during sensitive life stages and therefore warrant prioritization in cumulative exposure assessment. The combined application of targeted analysis and EOF in future studies of children's products would provide a more comprehensive characterization of PFAS exposure potential than either approach in isolation.

### Strengths and limitations of the study

A key strength of this study is the use of a saliva simulant to investigate PFAS migration under exposure conditions directly relevant to infant mouthing behavior, an exposure pathway that remains poorly characterized compared with dietary intake or dust ingestion. The compound-resolved analytical approach allowed identification of short-chain PFAS, particularly PFHxA, as major drivers of migration and enabled integration of experimental data with screening-level exposure metrics using MOE and reverse dosimetry. Several limitations should be acknowledged. Polymeric PFAS and total fluorine were not quantified, although total organofluorine in textiles can exceed extractable PFAS by several orders of magnitude [57, 61]. Toxicokinetic and toxicological data remain limited for many short-chain and emerging PFAS, requiring class-based assumptions that increase uncertainty in exposure estimates, particularly for oral exposure from textile-based products intended for children [58]. The experimental design also involved simplifications, including the use of artificial saliva, a limited number of products, and a single washing condition. Quantification relied on three isotopically labeled internal standards assigned on a structural class basis, an approach widely adopted in multi-residue PFAS analysis but which may not fully compensate for compound-specific matrix effects. However, the consistent composition of the artificial saliva matrix across samples and the recovery and duplicate performance reported in Section 2.7 are expected to limit this uncertainty. In addition, only oral exposure was considered, although dermal contact may also contribute to total exposure. Finally, differences between chronic population-based benchmarks (EFSA TWI) and screening-oriented MOE approaches may lead to divergent interpretations, especially when adult-derived guidance values are applied to infant exposure scenarios. Despite these limitations, this study provides early empirical evidence for an understudied exposure pathway and demonstrates the value of saliva-mediated migration testing as a screening tool to support prioritization of products and compounds in early-life PFAS exposure assessment.

## CONCLUSIONS

This study provides evidence that commercial stuffed toys can release PFAS into artificial saliva under mouthing-relevant conditions, supporting the plausibility of saliva-mediated oral exposure in early life, a pathway still poorly characterized [18, 19, 58]. PFAS were detected in all toys, with migration profiles dominated by short-chain PFCAs and with PFHxA as a key driver of inter-product variability, consistent with the higher mobility of short-chain PFAS and their increasing relevance as replacement substances [59, 62]. When translated into exposure metrics, estimated intakes for the four EFSA-regulated PFAS (PFOS, PFOA, PFHxS, PFNA) remained below the EFSA group TWI, indicating limited concern for these legacy PFAS under the assumptions applied [9]. However, MOE and reverse-dosimetry contextualization highlighted compound- and product-specific exposure relevance, particularly for short-chain and emerging PFAS such as PFHxA, for which toxicological and toxicokinetic data remain incomplete, increasing uncertainty and reinforcing the need for compound-specific data to support risk evaluation [54, 55, 62, 68]. The detection of 6:2 FTS in the majority of analysed products further suggests that fluorotelomer-based precursors may contribute to the short-chain PFCA burden through transformation processes, and represent a relevant aspect for future investigations of textile-based exposure pathways. Washing did not provide a uniformly protective effect: while most toys showed reduced ΣPFAS migration, a subset exhibited marked increases driven mainly by PFBA, indicating that simple mitigation measures may yield compound- and product-specific outcomes and should not be assumed effective without compound-resolved evaluation [18, 73, 74]. Overall, this work demonstrates the value of saliva-simulant migration testing coupled with screening-level exposure metrics as a complementary tool to identify exposure-relevant products and PFAS for prioritization in early-life exposure assessment and risk management [58, 82]. Future research would benefit from integrating compound-specific toxicokinetic data for short-chain PFAS, expanding biomonitoring in infants and toddlers, combining targeted PFAS analysis with non-targeted approaches such as extractable organofluorine, and combining oral, dermal, and aging-related exposure scenarios to better capture cumulative exposure during sensitive developmental windows.

## DATA AVAILABILITY

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request

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## AUTHOR CONTRIBUTIONS

LN conducted the experimental work, performed data analysis, processed the results, and contributed to manuscript drafting. JB and LS contributed to experimental work and analytical measurements. AO and LC conceived the study, secured funding, and coordinated the project. CR contributed to data analysis, supervised the scientific work, and verified the data. FL contributed to methodology development, supervised the experimental work, validated the data, and supported manuscript revision. DS conceived the study, supervised data processing, led manuscript writing, and provided scientific supervision.

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## COMPETING INTERESTS

The authors declare no competing interests.

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

No humans or animals were involved in this study. All methods were performed in accordance with the relevant guidelines and regulations.

## ADDITIONAL INFORMATION

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